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Commentary

International Consensus Meeting on Orthopaedic Infection: Differences Between ICM 2018 and ICM 2025

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ABSTRACT

Background: Periprosthetic joint infection (PJI) remains one of the most devastating complications after total joint arthroplasty. The International Consensus Meeting (ICM) has emerged as the leading global initiative to provide evidence-based recommendations on the prevention, diagnosis, and management of PJI. This study compares the recommendations of the 2018 and 2025 ICMs to assess how evolving evidence has shaped clinical guidance.
Methods: A total of 172 comparable questions were identified across thematic areas, including general, biofilm, hip and knee, shoulder, and spine.
Results: Changes in recommendations were observed in 37 questions (21.5%), while 33 (19.2%) were supported by stronger evidence in 2025 and 42 (24.4%) by weaker evidence. Notable evolution included recognition of the increased revision risk following total knee arthroplasty (TKA) after prior ipsilateral arthroscopy, the establishment of a minimum 12-week antibiotic regimen after debridement, antibiotics, and implant retention (DAIR) for acute PJI, and updated guidance on the role of repeat DAIR procedures. Beyond content, the 2025 ICM adopted more directive language with explicit grading of evidence and consensus strength, providing clearer clinical guidance.
Conclusions: These findings highlight how the rapid generation of high-quality evidence has transformed consensus recommendations, underscoring the central role of ICM in harmonizing global practice and defining research priorities in the management of PJI.
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Periprosthetic joint infections (PJIs) remain among the most devastating and unpredictable complications following total joint arthroplasty (TJA) [1]. They are associated with major morbidity and mortality, placing a multifactorial burden on patients, caregivers, surgeons, hospitals, health care systems, and economies. Furthermore, as the global number of TJAs continues to rise, driven by aging populations and expanded surgical indications, the absolute number of PJI cases is projected to increase substantially. Current evidence reports a PJI rate after primary joint arthroplasty to range between 0.5 and 2.3%, depending on the joint, population, and diagnostic criteria applied [2–5]. The impact of PJI extends far beyond the immediate surgical episode. Patients often require complex multi-stage revision procedures, prolonged antibiotic regimens, extended hospital stays, and intensive rehabilitation protocols [6,7]. Even when infection eradication is achieved, functional outcomes remain inferior to those of patients who do not have PJI, with persistent physical and psychosocial impairments widely documented [8,9]. Despite efforts, cure rates are

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Ethics Statement: This study is a narrative review and did not require ethical approval or informed consent.

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generally only 65 to 80% [10,11]. From a health economics perspective, the burden is equally substantial. Annual cost projections are expected to exceed $1.85 billion by 2030 [12,13].

In this context, consensus-building efforts are essential in providing standardized, evidence-based recommendations to guide clinical practice worldwide. However, as contemporary health care systems face increasing economic pressures and policy reforms mandate the elimination of nonevidence-based expenditures, Consensus Conferences, according to a modified Delphi method, have emerged as a critical instrument to synthesize all available knowledge, resolve areas of clinical uncertainty, and prioritize future research funding based on evidence gaps. The International Consensus Meeting (ICM) on Musculoskeletal Infection, first convened in 2013, represented a landmark initiative in orthopaedic surgery, focusing on the prevention, diagnosis, and management of PJI. The second ICM meeting was held in 2018.

Building on the success of these initial efforts, the third ICM, held in Istanbul in 2025, represented the most comprehensive and globally inclusive consensus to date regarding PJIs. Retaining the established Delphi methodology, the 2025 ICM integrated advanced digital tools, real-time data sharing, and artificial intelligence (AI)-assisted literature synthesis to ensure exhaustive evaluation of contemporary evidence. The meeting assembled over 1,000 international delegates, including orthopaedic surgeons, infectious disease specialists, microbiologists, pharmacologists, epidemiologists, and health care economists, who collaborated across thematic workgroups. More than 2,000 questions were submitted; after extensive review, 394 critical questions were selected for detailed discussion and consensus voting. Each was supported by explicit grading of the level of evidence and consensus strength. The present article aimed to explore whether there are differences in the recommendations provided for similar questions between the 2018 ICM and the 2025 ICM. Through this comparative analysis, we sought to highlight areas where evidence has evolved and identify shifts in clinical practice priorities, with a particular focus on analyzing the most clinically relevant questions.

Overview of the Changes in ICM 2025

A manual screening of all questions from the 2018 and 2025 ICM was performed. The objective of this process was to identify questions addressing similar clinical issues across both meetings. Through this screening, comparable questions were identified and subsequently grouped into predefined thematic categories: general session, biofilm, hip and knee, shoulder, and spine. All selected questions were then reevaluated in detail. For each pair of comparable questions, we systematically assessed potential differences in recommendations, level of evidence, and level of consensus. This approach allowed us to highlight changes that occurred over time in response to the evolving scientific literature and clinical experience. The process was conducted independently by multiple reviewers. No additional inclusion or exclusion criteria were applied, as the scope of the analysis was limited to questions explicitly formulated and voted in both the 2018 and 2025 ICM documents.

A total of 172 questions were identified between the 2018 and 2025 ICM. According to the thematic area, the majority of overlapping questions belonged to the general (37.8%) and hip and knee (36.6%) sessions, followed by shoulder (22.7%) and spine (2.9%) questions. A change in recommendation between the two ICMs was observed in 37 questions, corresponding to 21.5% of the total. Even when the recommendations remained unchanged, important shifts were noted in the underlying level of evidence and strength of consensus. Specifically, 33 questions (19.2%) were supported by stronger evidence in 2025 compared with 2018, while 42 questions (24.4%) were associated with weaker evidence.

Emergence of High-Quality Evidence Between 2018 and 2025

Between the release of the ICM in 2018 and its landmark update in 2025, the landscape of PJI management has undergone a profound transformation, driven by the emergence of high-quality evidence across clinical and basic science (biofilms, etc.) domains. Perhaps most notably, the advent of large-scale, multicenter prospective cohort studies, including the international trials, provided unprecedented clarity on outcomes associated with revision risk based on the timing of knee arthroscopy before total knee arthroplasty (TKA). For the ICM 2025, the results of new large cohort studies and timing analyses in 2020 and 2021 have shown an increased risk of revision when TKA is performed soon after arthroscopy, demonstrating a clearer association [14–17]. Additionally, a database analysis in 2023 confirmed the increased risk of postoperative complications in patients undergoing ipsilateral TKA within one year of TKA [18]. These findings have been supported by systematic reviews and meta-analyses, which concluded that prior arthroscopy may negatively affect TKA outcomes [19,20]. Together, this evidence explains the shift in the 2025 ICM recommendation, from “no evidence” to “yes, increased the risk”. There is a clinically meaningful increased risk of postoperative infection following TKA in patients who have a history of ipsilateral knee arthroscopy within 6 months.’ [17].

In addition, new high-quality clinical findings are also filling in some of the blank recommendations. For example, the recommendation of the optimal duration of antimicrobial treatment for patients undergoing debridement, antibiotics, and implant retention (DAIR) for acute PJI was “unknown” in ICM 2018. Due to the new clinical evidence published between 2018 and 2025, the recommendation is now “a minimum of 12 weeks of antibiotic therapy is recommended for patients undergoing DAIR for acute PJI.” [21] Based on a randomized, open-label, noninferiority study published in the New England Journal of Medicine in 2021 that directly addressed the optimal duration of antimicrobial therapy for PJIs treated with different surgical interventions, including DAIR, the optimal duration of antimicrobial treatment for patients who have acute bacterial PJI managed with DAIR is 12 weeks [22].

New evidence also induces opposite changes in some questions. The recommendation of the question “is there a role for repeat DAIR in patients who have acute PJI of hip or knee?” in ICM 2025, which is similar to the question about the recommendation of maximum times of DAIR procedure(s) before removal of components in ICM 2018, is developed to “possibly. Repeat DAIR appears to increase the overall success rate of infection eradication when performed after failure of initial DAIR” from “after one failed DAIR procedure, strong consideration should be given to removal of components.” [21] Compared with the 2018 version, this new recommendation also includes data from six clinical studies published between 2018 and 2024 and further derives new recommendations based on systematic reviews and meta-analyses. The meta-analysis indicated that the success rate for a single DAIR was slightly higher than the success rate observed for repeat DAIR, although this difference was not statistically significant. Nevertheless, when outcomes from both single and repeat DAIR procedures were combined, the overall success rate markedly increased compared to the single DAIR approach. These findings suggest that at least half of the patients undergoing resection arthroplasty for persistent acute infections might be treated with implant retention.

In essence, the divergence between the two consensus documents reflects the cumulative power of new evidence. The 2025 consensus stands as new evidence of how rapidly evolving science

can reshape PJI practice. The emergence of evidence in the interim did not merely inform updates; it can modify the clinical decision-making that rests in PJIs.

Changes in the Wording of Recommendations

An important difference between the 2018 and 2025 ICM resides not only in the content of the recommendations but in their wording. The 2018 consensus frequently included cautious and descriptive language, such as "may be considered" or "insufficient evidence to support," which reflected the limited data available at that time. In contrast, the 2025 ICM often employed more assertive and practical language, supported by explicit grading of evidence and consensus strength. For example, in 2018, the recommendation for the optimal antibiotic prophylaxis in patients who had known anaphylactic penicillin allergies stated that "cephalosporins for patients with anaphylactic penicillin allergies may be given following skin testing." By 2025, this wording was reformulated into a clear directive with a strong recommendation: "It is safe and appropriate to administer a first-generation cephalosporin (cefazolin) as first-line antibiotic prophylaxis in the operating room setting for patients who have a documented penicillin allergy" [23], thus reducing ambiguity and offering stronger clinical guidance. A similar shift occurred with one-stage exchange in culture-negative prosthetic joint infections (PJIs): in 2018, it was suggested that "this approach remains a plausible option for the management of PJIs in a selected group of individuals," whereas in 2025, the statement was rephrased positively, affirming that "yes, one-stage exchange can be considered a viable option for patients with culture-negative PJI" [24]. Another striking example is prior corticosteroid injection for presumed PJI after shoulder arthroplasty: in 2018, the guidance was limited to noting "may increase the risk for surgical site infection or shoulder PJI", while in 2025 it became a firm directive: "Corticosteroid injection within 3 months of shoulder arthroplasty has been associated with an increased risk of PJI" [25]. These shifts illustrate how the 2025 ICM moved from permissive or vague expressions toward more directive ones, enhancing clarity and applicability for clinical decision-making.

More Diverse and International Delegates

Some changes to the recommendation may stem from our deliberately expanded, globally diverse delegates. While the 2018 consensus involved 880 delegates from 94 countries and regions (27.3% from the US), the 2025 effort engaged 1,205 delegates representing 102 countries and regions, including enhanced participation from Asia and Africa. Notably, the US representation had reduced from 43% in 2018 to 24% in 2025. This intentional geographical diversification enriched perspectives. However, voting patterns from different countries and regions may reflect a stronger emphasis on cost-effective diagnostics (e.g., C-Reactive Protein versus other novel, but expensive biomarkers) and one- or two-stage surgical protocols suitable for resource-limited regions. Moreover, divergent national health care policies may potentially correlate with voting tendencies of delegates.

Conclusions

The number of publications related to PJI has been increasing exponentially year after year. Due to the soul-crushing gravity of orthopedic infections, the scientific community has engaged in meaningful research to address this issue for our patients. However, despite these admirable efforts, there are many uncertainties related to the management of patients who have orthopedic infections. Generating evidence for some of the issues requires immense logistical and financial commitments. Some of the prospective studies on the subject have taken years to conclude. Recognizing the latter challenge, the International Consensus Meeting has been gathering experts from around the world to discuss and deliberate on some of the questions related to orthopedic infections. The purpose of these meetings is to provide recommendations, based on evidence whenever possible, or reach consensus on issues that lack adequate evidence. The goal of the meeting is to bring everyone to sing from the same hymnbook and also create a list of research priorities. The areas identified as "in need of research" in 2018 led to the conduct of large trials and the generation of evidence that fueled the evidence-based recommendations of 2025. The meeting was a resounding success. The delegates dedicated nearly two years of their time to conduct the comprehensive literature review, traveled at their own expense to Istanbul to sit through 8-hour meetings for three days, which is a testimony to the selfless dedication of the medical profession. We should be proud.

CRediT authorship contribution statement

Augusto Ferrini: Writing – original draft, Formal analysis, Conceptualization. **Mincong He:** Writing – original draft, Formal analysis. **Martí Bernaus:** Supervision. **Margarita Veloso:** Supervision. **Michael A. Mont:** Supervision. **Javad Parvizi:** Writing – review & editing, Supervision.

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